

## Monolingual Results

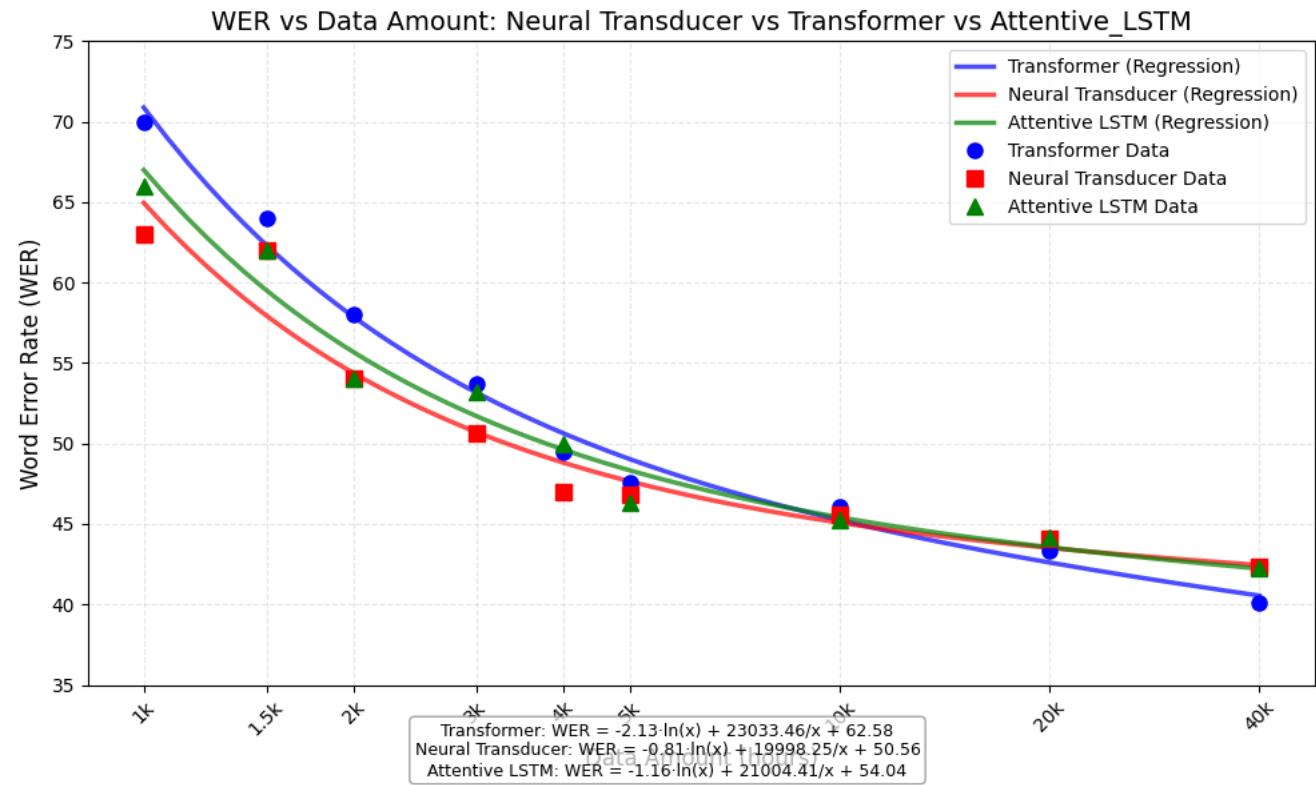
| Language               | Pair n-Gram Model<br>(baseline 2020) | Encoder-decoder Transformer<br>(baseline 2020) | Neural Transducer<br>(baseline 2021/22) | Attentive_LSTM<br>(baseline 2024) |
|------------------------|--------------------------------------|------------------------------------------------|-----------------------------------------|-----------------------------------|
| Adyghe                 | 27.00                                | 39.00                                          | 20.00                                   | 30.00                             |
| Arabic                 | 43.00                                | 43.00                                          | 53.00                                   | 45.00                             |
| Armenian (Eastern)     | 17.00                                | 16.00                                          | 15.00                                   | 20.00                             |
| Assamese               | 14.00                                | 15.00                                          | 7.00                                    | 12.00                             |
| Belarusian             | 7.00                                 | 6.00                                           | 2.00                                    | 2.00                              |
| Bengali                | 70.00                                | 78.00                                          | 68.00                                   | 67.00                             |
| Bulgarian              | 37.00                                | 30.00                                          | 32.00                                   | 27.00                             |
| Burmese                | 39.00                                | 41.00                                          | 29.00                                   | 35.00                             |
| Cebuano                | 22.00                                | 26.00                                          | 20.00                                   | 20.00                             |
| Central Khmer          | 56.00                                | 43.00                                          | 31.00                                   | 35.00                             |
| Dutch                  | 32.00                                | 29.00                                          | 24.00                                   | 23.00                             |
| Eastern Lawa           | 44.00                                | 24.00                                          | 8.00                                    | 12.00                             |
| English                | 70.00                                | 70.00                                          | 63.00                                   | 66.00                             |
| French                 | 31.00                                | 29.00                                          | 23.00                                   | 27.00                             |
| Georgian               | 0.00                                 | 4.00                                           | 0.00                                    | 1.00                              |
| German                 | 49.00                                | 56.00                                          | 46.00                                   | 45.00                             |
| Greek                  | 27.00                                | 26.00                                          | 20.00                                   | 27.00                             |
| Hindi                  | 23.00                                | 24.00                                          | 11.00                                   | 13.00                             |
| Hungarian              | 9.00                                 | 11.00                                          | 7.00                                    | 8.00                              |
| Icelandic              | 35.00                                | 24.00                                          | 12.00                                   | 16.00                             |
| Indonesian             | 58.00                                | 52.00                                          | 64.00                                   | 73.00                             |
| Irish                  | 57.00                                | 46.00                                          | 43.00                                   | 39.00                             |
| Italian                | 22.00                                | 21.00                                          | 15.00                                   | 16.00                             |
| Japanese (Hiragana)    | 23.00                                | 20.00                                          | 10.00                                   | 12.00                             |
| Korean                 | 81.00                                | 89.00                                          | 23.00                                   | 100.00                            |
| Latvian                | 51.00                                | 51.00                                          | 51.00                                   | 50.00                             |
| Lithuanian             | 32.00                                | 35.00                                          | 33.00                                   | 31.00                             |
| Macedonian             | 6.00                                 | 5.00                                           | 5.00                                    | 5.00                              |
| Maltese (Latin)        | 27.00                                | 24.00                                          | 17.00                                   | 22.00                             |
| Norwegian Nynorsk      | 61.00                                | 69.00                                          | 66.00                                   | 69.00                             |
| Pashto                 | 70.00                                | 68.00                                          | 67.00                                   | 66.00                             |
| Persian (Classical)    | 51.00                                | 58.00                                          | 57.00                                   | 51.00                             |
| Persian (Iranian)      | 66.00                                | 63.00                                          | 65.00                                   | 65.00                             |
| Polish                 | 9.00                                 | 8.00                                           | 4.00                                    | 7.00                              |
| Romanian               | 10.00                                | 17.00                                          | 9.00                                    | 17.00                             |
| Russian                | 31.00                                | 21.00                                          | 23.00                                   | 25.00                             |
| Serbo-Croatian (Latin) | 84.00                                | 69.00                                          | 64.00                                   | 69.00                             |
| Shan                   | 6.00                                 | 10.00                                          | 5.00                                    | 4.00                              |
| Slovenian              | 73.00                                | 52.00                                          | 56.00                                   | 50.00                             |
| Spanish                | 3.00                                 | 10.00                                          | 4.00                                    | 3.00                              |
| Swedish                | 67.00                                | 68.00                                          | 59.00                                   | 61.00                             |
| Tagalog                | 11.00                                | 17.00                                          | 13.00                                   | 11.00                             |

| Language                 | Pair n-Gram Model<br>(baseline 2020) | Encoder-decoder Transformer<br>(baseline 2020) | Neural Transducer<br>(baseline 2021/22) | Attentive_LSTM<br>(baseline 2024) |
|--------------------------|--------------------------------------|------------------------------------------------|-----------------------------------------|-----------------------------------|
| Thai                     | 70.00                                | 49.00                                          | 39.00                                   | 42.00                             |
| Ukrainian                | 27.00                                | 26.00                                          | 19.00                                   | 18.00                             |
| Urdu                     | 67.00                                | 66.00                                          | 72.00                                   | 77.00                             |
| Uyghur                   | 0.00                                 | 3.00                                           | 0.00                                    | 2.00                              |
| Vietnamese (Hanoi)       | 44.00                                | 20.00                                          | 5.00                                    | 4.00                              |
| Welsh                    | 33.00                                | 20.00                                          | 12.00                                   | 19.00                             |
| Welsh (Southern Dialect) | 28.00                                | 18.00                                          | 13.00                                   | 13.00                             |
| Macro-average WER        | 37.14                                | 34.88                                          | 28.65                                   | 31.67                             |

## How Data Amount Affects Performance of Transformer Model

Comparing best performing model and transformer on different amount fo English data

| Data Amount | Neutral Transducer | Encoder-Decoder Transformer | Attentive LSTM |
|-------------|--------------------|-----------------------------|----------------|
| 1k          | 63.00              | 70.00                       | 66.00          |
| 1.5k        | 62.00              | 64.00                       | 62.00          |
| 2k          | 54.00              | 58.00                       | 54.00          |
| 3k          | 50.67              | 53.67                       | 53.20          |
| 4k          | 47.00              | 49.50                       | 50.00          |
| 5k          | 46.80              | 47.60                       | 46.30          |
| 10k         | 45.60              | 46.10                       | 45.21          |
| 20k         | 44.05              | 43.35                       | 44.13          |
| 40k         | 42.38              | 40.10                       | 42.23          |



## Multilingual Approach

A tag comprising a language code (e.g. <[ENG]>, <[GER]>) is prepended to each grapheme sequence source. The datasets of each language are simply concatenated.

Does converting the language tag to the corresponding script of the language improve the model performance?

Using `slavic_cyrillic` dataset we got following validation WER:

| Tag in             | Neural Transducer | Encode-Decoder Transformer | Attentive_LSTM |
|--------------------|-------------------|----------------------------|----------------|
| Uppercase Cyrillic | 39.67             | 28.00                      | 28.00          |
| Uppercase Latin    | 31.00             | 28.00                      | 27.00          |
| Lowercase Latin    | 32.33             | 28.00                      | 27.00          |
| No tag             | 45.00             | 44.67                      | 44.67          |

Results

Same Language Family and Same Script

|              |          |                        | Neural Transducer |             | Encoder-decoder Transformer |             | Attentive_LSTM |             |
|--------------|----------|------------------------|-------------------|-------------|-----------------------------|-------------|----------------|-------------|
| Family       | Script   | Language               | Multilingual      | Monolingual | Multilingual                | Monolingual | Multilingual   | Monolingual |
| Germanic     | Latin    | German                 | 48                | 46          | 47                          | 56          | 42             | 45          |
|              |          | Dutch                  | 25                | 24          | 21                          | 29          | 20             | 23          |
|              |          | Swedish                | 62                | 59          | 60                          | 68          | 60             | 61          |
| Romance      | Latin    | Italian                | 26                | 15          | 22                          | 21          | 18             | 16          |
|              |          | Spanish                | 12                | 4           | 3                           | 10          | 2              | 3           |
|              |          | Romanian               | 18                | 9           | 13                          | 17          | 13             | 17          |
| Slavic       | Cyrillic | Russian                | 27                | 23          | 18                          | 21          | 22             | 25          |
|              |          | Ukrainian              | 22                | 19          | 27                          | 26          | 19             | 18          |
|              |          | Bulgarian              | 35                | 32          | 27                          | 30          | 31             | 27          |
| Slavic       | Latin    | Serbo-Croatian (Latin) | 67                | 64          | 63                          | 69          | 68             | 69          |
|              |          | Polish                 | 13                | 4           | 9                           | 8           | 9              | 7           |
|              |          | Slovenian              | 59                | 56          | 52                          | 52          | 50             | 50          |
| Indo-Iranian | Arabic   | Pashto                 | 79                | 67          | 66                          | 68          | 64             | 66          |
|              |          | Persian (Iranian)      | 73                | 65          | 67                          | 63          | 64             | 65          |
|              |          | Urdu                   | 78                | 72          | 64                          | 66          | 71             | 77          |
| Austronesian | Latin    | Indonesian             | 66                | 64          | 57                          | 52          | 55             | 73          |
|              |          | Cebuano                | 26                | 13          | 24                          | 26          | 22             | 20          |
|              |          | Tagalog                | 15                | 20          | 18                          | 17          | 16             | 11          |

Different Language Family and Same Script

|               |        |                   | Neural Transducer |             | Encoder-decoder Transformer |             | Attentive_LSTM |             |
|---------------|--------|-------------------|-------------------|-------------|-----------------------------|-------------|----------------|-------------|
| Family        | Script | Language          | Multilingual      | Monolingual | Multilingual                | Monolingual | Multilingual   | Monolingual |
| Afro-Asiatic  | Arabic | Arabic            | 57                | 53          | 38                          | 43          | 53             | 45          |
| Indo-European |        | Persian (Classic) | 62                | 57          | 45                          | 58          | 55             | 51          |
| Turkic        |        | Uyghur            | 8                 | 0           | 2                           | 3           | 4              | 2           |
| Indo-European | Latin  | Italian           | 19                | 15          | 20                          | 21          | 17             | 16          |
| Afro-Asiatic  |        | Maltese (Latin)   | 18                | 17          | 21                          | 24          | 15             | 22          |
| Uralic        |        | Hungarian         | 6                 | 7           | 8                           | 11          | 3              | 8           |

Same Language Family and Different Script

|               |          |                    | Neural Transducer |             | Encoder-decoder Transformer |             | Attentive_LSTM |             |
|---------------|----------|--------------------|-------------------|-------------|-----------------------------|-------------|----------------|-------------|
| Family        | Script   | Language           | Multilingual      | Monolingual | Multilingual                | Monolingual | Multilingual   | Monolingual |
| Indo-European | Armenian | Armenian (Eastern) | 20                | 15          | 14                          | 16          | 18             | 20          |
|               | Greek    | Greek              | 21                | 20          | 25                          | 26          | 23             | 27          |

|          |         | Neural Transducer |    | Encoder-decoder Transformer |    | Attentive_LSTM |    |
|----------|---------|-------------------|----|-----------------------------|----|----------------|----|
| Cyrillic | Russian | 22                | 23 | 21                          | 21 | 20             | 25 |

## Reference Result from previous shared task

2022

Model

**Baseline:** A neural transducer system using an imitation learning paradigm (dyNET framework)

**Submissions:**

1.

Tü-G2P: A series of sequence labelling systems to G2P tasks, which use simpler alignment rather than dynamic transducer-based alignment.(Pytorch)
2.

Hammond (Repo): A non-neural system based on OpenFST and uses weighted finite-state transducers and expectation-maximization to compute the best many-to-many alignment of letters and phonetic symbol
3.

mSLAM: Non-archival; abstract only; useless
4.

NFST: Non-archival; abstract only; useless

Reference Result:

- Baseline

| Language | Bengali | Burmese | German | Irish | Italian | Persian | Swedish | Tagalog | Thai  | Ukrainian | Macro-average |
|----------|---------|---------|--------|-------|---------|---------|---------|---------|-------|-----------|---------------|
| WER      | 67.12   | 29.00   | 42.00  | 38.00 | 15.00   | 59.65   | 45.00   | 20.00   | 21.00 | 32.00     | 36.88         |

- Hammond (trigram alignment)

| Language | Bengali | Burmese | German | Irish | Italian | Persian | Swedish | Tagalog | Thai  | Ukrainian | Macro-average |
|----------|---------|---------|--------|-------|---------|---------|---------|---------|-------|-----------|---------------|
| WER      | 68.49   | 48.00   | 61.00  | 51.00 | 25.00   | 67.86   | 55.00   | 18.00   | 72.00 | 50.00     | 51.63         |

2024

Model

**Baseline:** attentive\_gru, attentive\_lstm, gru, hard\_attention\_gru, lstm, pointer\_generator\_gru, transducer\_gru, transformer(20\40\60 epochs)

Reference Result:

- Baseline

| Models  | gru   | lstm  | attentive_gru | attentive_lstm | hard_attention_gru | hard_attention_gru (Arab) | pointer_generator_gru | transducer_gru | transforme |
|---------|-------|-------|---------------|----------------|--------------------|---------------------------|-----------------------|----------------|------------|
| WER (%) | 43.75 | 44.25 | 63.08         | 47.25          | 40.67              | 31.33                     | 62.17                 | 69.33          | 78.25      |

- Best performance model on all datasets(hard\_attenton\_gru)

| Languages | Arabic | Bulgarian | English | Persian | Indonesian | Macedonian | Pashto | Russian | Spanish | Tagalog | Ukrainian | Urdu  |
|-----------|--------|-----------|---------|---------|------------|------------|--------|---------|---------|---------|-----------|-------|
| WER       | 31.33  | 20.00     | 58.00   | 29.67   | 55.33      | 3.67       | 44.33  | 10.33   | 5.00    | 40.33   | 15.67     | 64.00 |

- GRU, LSTM, Transformer on different languages

| Model/Languages | English | Pashto | Russian | Spanish |
|-----------------|---------|--------|---------|---------|
| GRU             | 31.00   | 39.00  | 14.00   | 9.00    |
| LSTM            | 48.33   | 57.67  | 10.33   | 9.00    |
| Transformer     | 81.33   | 77.00  | 35.67   | 24.67   |

- Comparison between hard\_attention\_gru, attentive\_lstm

| Model / Language   | Korean | Bengali | Indonesian | Pashto | Swedish |
|--------------------|--------|---------|------------|--------|---------|
| hard_attention_gru | 98.00  | 87.00   | 85.00      | 78.00  | 66.00   |
| attentive_lstm     | 100.00 | 67.00   | 73.00      | 66.00  | 61.00   |

- Comparison between attentive\_gru and attentive\_lstm

The comparison of validation accuracy (val\_accuracy) between attentive\_gru (GRU) and attentive\_lstm (LSTM) across all languages shows:

- LSTM val\_accuracy higher than GRU: 20 times
- LSTM val\_accuracy lower than GRU: 22 times
- LSTM val\_accuracy equal to GRU: 6 times
- Average LSTM val\_accuracy: 0.6987
- Average GRU val\_accuracy: 0.7049

This indicates that, on average, attentive\_gru achieves slightly higher validation accuracy than attentive\_lstm, although the performance varies across different languages.

- attentive\_lstm performance on Adyghe and Bengali with different parameters(DEV\_WER)

| Params                                | Adyghe WER | Bengali WER | Dutch WER | Urdu WER |
|---------------------------------------|------------|-------------|-----------|----------|
| default                               | 31.00      | 65.00       | 23.00     | 77.00    |
| 4 encoder_layers + 1 decoder_layer    | 51.00      | 70.00       | 45.00     | 80.00    |
| 256 embedding_size + 1024 hidden_size | 37.00      | 67.00       | 30.00     | 78.00    |
| 0.1 label_smoothing                   | 33.00      | 65.00       | 21.00     | 75.00    |